

Stage 2 Specialist Mathematics

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Subject Co-ordinator	R Scott
Teaching Staff	M Durant, R Scott
Texts	<i>Mathematics for Australia- 12 Specialist Mathematics; Haese et al</i>
Lesson Allocation	Five 45 or 50 minute lessons per week

Rationale

Specialist Mathematics draws on and deepens students' mathematical knowledge, skills, and understanding, and provides opportunities for students to develop their skills in using rigorous mathematical arguments and proofs, and using mathematical models. It includes the study of functions and calculus.

The subject leads to study in a range of tertiary courses such as mathematical sciences, engineering, computer science, and physical sciences. Students envisaging careers in related fields will benefit from studying this subject.

Specialist Mathematics is designed to be studied in conjunction with Mathematical Methods.

Learning Requirements

The learning requirements summarise the key skills, knowledge and understanding that students are expected to develop and demonstrate through learning in Stage 2 Mathematical Methods.

In this subject, students are expected to:

1. understand mathematical concepts, demonstrate mathematical skills, and apply mathematical techniques
2. investigate and analyse mathematical information in a variety of contexts
3. think mathematically by posing questions, solving problems, applying models, and making, testing, and proving conjectures
4. interpret results, draw conclusions, and determine the reasonableness of solutions in context
5. make discerning use of electronic technology to solve problems and to refine and extend mathematical knowledge
6. communicate mathematically and present mathematical information in a variety of ways.

These learning requirements form the basis of the:

- learning scope
- evidence of learning that students provide
- assessment design criteria
- levels of achievement described in the performance standards.

Term 1	Content	Text Reference	Assessment
Week 1 Mon: Staff PD Tues: Year 11-12 activities Wed: TBC lessons commence L3.	Mathematical Induction	1B (not Q4,6,7, 8,11c,13)	
Week 2 Mon: Commencement Ceremony Wed: SS Photos	Induction continued. Polynomials: Operations, Roots, zeros and factors	2A, B	
Week 3 Wed - Thurs: 11 IB CAS interviews	Real Polynomials: Equality, Equating coefficients, Definition, $P = dQ + R$, Division, Synthetic Division, The Remainder theorem	2C, D, E	
Week 4 Sun - Wed: Yr12 geography trip	The Factor theorem, Factorisation of $x^n \pm a^n$, The Fundamental theorem of Algebra	2F, G	
Week 5	Graphing real polynomials, Polynomial Equations	2I (briefly), J	SAT 1: Induction and Polynomials No calculator No notes
Week 6 Wed: SMC Thurs: 12 IB CAS interviews	Complex Numbers: Complex Plane, Modulus and Argument,	"Exploration of solutions to $z^n = w$ " "Operations with $\mathbb{C}$ " Worksheet 4A, 4B & Exercise on p107 of electronic text, C1	
Week 7 Mon: PH Adelaide Cup Day Thurs: L3 & 4 Guest speaker Sat: Head of the River Fri-Sun: Pedal Prix	Polar form, argument, Multiplying and dividing in Polar form, argument properties, Conversion between Cartesian and Polar form, Locus problems	4C2, C3, 4D (ext)	
Week 8 Sat: SS Formal	De Moivre's Theorem, Solving $z^n = C$, the n^{th} roots of unity,	4E, 4F	
Week 9 Tues: Summer sport photos Tues: 11 CAS interviews Wed - Thurs: 11 Tourism Trip Thurs: Swimming Carnival TBC Fri: Good Friday	Catch up & revision Functions: Composites, Inverse functions	3A, 3B	SAT 2: Complex Numbers
EASTER			
Week 10 Mon: Easter Monday Wed- Sat: Mt Gambier Fieldtrip	Reciprocal and rational functions	3C, 3D, 3E + "Graphs of Rational Functions" Worksheet	
Week 11 Wed - Sun: Melbourne football trip	Absolute value functions (NB: Text does not cover $ f(x) $.)	"Modulus Functions" worksheet 3F	

Term 2	Content	Text Reference	Assessment
Week 1 Mon: Staff PD day Tues – Fri: 12 IB Mini-exams Fri, Sat, Sun: Generations in Jazz	3D Vectors: Vectors in space, Operations with Vectors, Vector algebra, Coordinate geometry of vectors, Parallelism,	5A, 5B, 5C 5D, 5E	
Week 2 Tues-Fri: 12 Tourism to Cairns	Collinearity and unit vectors Scalar product and algebraic properties, Scalar product geometric properties, Vector proof Vector product definition and properties	5F, 5G, 5H + "Vector Projection" & "Vector Proof Suppl Qns" 5I	
Week 3 Tues: AMC Wed-Thurs: 11 Tourism to Barossa Wed- Fri: 12 Art to Melbourne Fri -Sat: Pedal Prix	Integration: Integration Rules Inverse Trig Integrals, Integration by Substitution	7A 7B, C1 (not Q10,11 ext) C2	SAT 3: Functions and Vectors I
Week 4 Thurs: 11 Retreat Day	Integration by Parts, Areas, Solids of revolution	7D, E 7F	
Week 5 Fri: 11 IB study day	Catchup & Test Rates of Change and D.E's: Implicit Differentiation, Related rates	8A, B	SAT 4: Integration, Related Rates and Implicit Differentiation
Week 6 11 IB exams week Fri: 11 SACE Study Day	Differential Equations Separable D.E's	8C (Q1 &2) 8D, 8E	
Week 7 Mon: PH King's B'Day Tues: 11 IB classes resume. Tues-Fri: 11 SACE exams	Slope fields, Problem-solving	<i>"Intro to Slope Fields" wksht</i> + 8F, G,	
Week 8 Mon- Tues: more 11 SACE exams Wed- SACE classes resume Sat: UK cricket tour depart	Logistic growth, Parametric equations	8H, 9A	
Week 9 M-F: Yr11 Work Experience Week (IB students from Wednesday) Fri: last day of term (full day)	Investigation		Investigation Start

Term 3	Content	Text Reference	Assessment
Week 1 Mon: Students return	Pairs of uniformly varying quantities, Pairs of non-uniformly varying quantities	9B, 9C	
Week 2 Mon-Wed: 11 IB Theatre & SACE Drama, Theatre Intensive Week Fri- Sun: Pedal Prix	Bezier Curves, Circular and Elliptical Motion, Arc length	9D (Q2-4), 9E + extra, 9F	
Week 3 Tues: SS Winter Sport Photos Wed L3/4: Yr11 RAP presentation Thurs: 11 CAS interviews Fri: 12 Valette Photos	Investigation		Draft Investigation
Week 4 Wed, Thu, Fri: ABODA music festival Fri- Mon(wk5): Pedal Prix	Review + Test Vectors: Lines and Planes Area, Lines in 2D and 3D	6A, B	SAT 5: DE's and Rates of Change
Week 5	Angle between two lines, Constant velocity problems, Distance of a point to a line, Intersecting lines, Relationships between lines,	6C, D, E 6F, 6G	
Week 6 Mon, Thurs-Fri: 12 IB CAS interviews Thurs- Fri: Music Showcase	Cartesian Eqn of a plane	6H	
Week 7 Mon: 12 IB study day Tues-Th: 12 IB mock exams Friday: Student Free Day	Angles in Space, Row operations, Intersecting planes	6I, J, K	
Week 8 M-Th: 12 IB mock exams Fri: 12 IB classes resume M-F: 12 SACE mock exams	Mock Exams		
Week 9 Mon: 12 SACE classes resume Tues: MS/SS Music Photos Fri-Sun: Pedal Prix	Investigation Final + Submit		Investigation Final Due
Week 10 Tues afternoon: House athletics pre-events Wed: IGSSA Athletics Day Thurs: Achilles Cup Fri: full day to finish	Catch-up, revision and test		SAT 6: Lines & Planes

Term 4	Content	Text Reference	Assessment
Week 1 Mon- Tues: 11 IB CAS interviews Wed: House Athletics Day Thurs: Yr12 leavers activities & valedictory rehearsals Fri: Valedictory Ceremony	Leavers Week		
Week 2 12 IB exams commence	Swot Vac		
Week 3 Week 4+ 12 SACE exams commence	Swot Vac Exams		

ASSESSMENT

The mark required for SACE assessment will be calculated as follows:

Assessment type	Number of tasks	Total Weighting
Skills and Applications Tasks	SAT 1: Induction & Polynomials – no notes nor calculator SAT 2: Complex Numbers – 1 A4 page of notes and calculator permitted SAT 3: Functions and Graphs & Vectors I – 1 A4 page of notes and calculator permitted SAT 4: Integration, Implicit Differentiation and Related Rates – 1 A4 page of notes and calculator permitted SAT 5: Vector Calculus – 1 A4 page of notes and calculator permitted SAT 6: Vectors II: Lines and Planes – 1 A4 page of notes and calculator permitted	50%
Mathematical Investigation	1 task	20%
Examination (130 minutes)	External Two double sided A4 notes pages and calculator permitted	30%

- The six tests will be approximately equal in weighting.
- The mock examination in Term 3 will be for Pembroke assessment and will aid in the calculation of the predicted score.